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Polyetheretherketone (PEEK): A Comprehensive Review of Advanced Polymer Engineering, Biomedical Applications, and Emerging Technological Frontiers

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Keywords

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Abstract

Polyetheretherketone (PEEK) is a top-level polymer combining cutting-edge engineering with new medicine. PEEK is noted for its superior mechanical properties, thermal stability (Tg = 143°C, Tm ≈ 343°C), and chemical resistance against corrosive agents, and its role in green manufacturing, setting new standards in several fields, from aerospace to medicine. This article examines new methods of producing PEEK, more crosslinks, and nucleophilic polycondensation, which augmented its thermal stability and strength. Fused filament fabrication (FFF) and digital light processing (DLP) are new methods of producing tailor-made implants for patients and lightweight aircraft components, illustrating how versatile PEEK is in a 3D printing process. PEEK (polyether ether ketone) is beneficial in biomedical applications because its stiffness matches that of human bone (3–4 GPa), reducing stress shielding in implants. While it is bioinert, recent surface treatments like plasma grafting and hydroxyapatite coating improve its integration with bone. Outside of the medical field, PEEK is a long-lasting lithium-ion battery separator and is suitable for water applications due to its resistance to seawater wear. Microfluidic systems that take advantage of PEEK's inert qualities and radiation-shielded composites like boron carbide-PEEK for nuclear applications are further developments. Despite its versatility, PEEK has issues with recycling, durability in the face of intense radiation, and standardization in 3D-printed medical devices. Interdisciplinary cooperation is essential for developing PEEK in green manufacturing and technology in order to overcome these challenges.

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1. INTRODUCTION

Polyetheretherketone (PEEK) is a high-performance thermoplastic, a new industrial benchmark in material-based industries such as aerospace, biomedicine, and energy. Its combined chemical resistance, thermal stability, and mechanics make PEEK distinct from conventional materials and distinguish it in applications with stresses and durability. Its alternative to metals is that PEEK's modulus of elasticity (very similar to that of a bone) minimises bone loss, where PEEK is quite suitable for applications in orthopaedics and dentistry (Dondani et al., 2023). Further, its excellent chemical resistance allows PEEK to function in hostile environments in aerospace, where tolerance of the material under severe conditions creates superior fuel efficiency and operational life (Kurtz & Devine, 2007).

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Since its availability in the market in the 1980s with its diverse molecular structure of alternating ether and ketone bonds, PEEK has enabled new processes such as additive manufacturing. This enables the manufacture of patient-specific devices with intricate geometric configurations, leading to increased personalisation and efficiency of biomedical devices (Pulipaka et al., 2023).With markets globally, and predominantly in the medical sector, expected to grow at a compound annual growth rate (CAGR) of 6.7% by 2030, the projected market growth is summarized in Table 1 (Research, 2024). This table presents the global medical PEEK market valuation from 2024 to 2033, illustrating the increasing demand across sectors. Demand for PEEK will continue to climb, particularly in orthopaedic and dental devices and aerospace and energy applications.

Future breakthroughs in composite design and 3D printing will continue to widen the functionality of PEEK and enable still more sophisticated and sustainable applications. As with every hurdle we face in trying to keep up with the demands of modern engineering, increased presence in sustainable processes and biomedical applications foretells the material's role to continue to be substantial. As surface treatment and recycling technologies continue to evolve, PEEK will be a foundation material for existing industries and next-generation sustainable technologies.

The flexibility of the material is due to molecular make-up, with alternating ketone (-C(=O)-) and ether (-O-) bonds, allowing the material to be easily worked and yet be useful. In biomedical applications, PEEK's modulus of elasticity (3–4 GPa) closely matches that of cortical bone of humans and obviates the effects of metal implantation-induced stress-shielding effects (Dondani et al., 2023). However, its bioinertness produces osseointegration problems, and much effort has been spent on surface modification and hydroxyapatite, carbon fibre and other functional filler-reinforced composite grades (Chen et al., 2022). Parallel to this have been breakthroughs in additive manufacturing enabling precise patient-specific implant manufacture with complex geometry to continue to expand PEEK's clinical applications (Pulipaka et al., 2023).

The review considers three fundamental issues: (1) How have composite science and polymer development improved PEEK's functional efficiency? (2) How has PEEK increased industrial and biomedical applications with the help of advances in technology? (3) What are the unresolved challenges in PEEK's long-term operation under harsh conditions? This review encompasses peer-reviewed literature over the last decade, focusing on those with robust experimental proof.

The purpose of this review is to present a thorough overview of the uses of PEEK in all types of fields. Section 2 will cover how PEEK is made as well as what PEEK is made of and how these affect the overall properties of PEEK; i.e., what it can withstand when placed under various stresses/strains and how it will react with other materials. Moreover, Sections 3–6 will investigate the physical/mechanical properties of PEEK; specifically, thermal stability, strength and durability/frictional wear resistance.

In addition, Section 4 will provide detailed information regarding chemico-environmental resistance of PEEK and how well it performs when subjected to various chemical/physical/biological conditions. Section 5 will

summarize current advancements and applications of PEEK in a variety of industries (biomedicine – orthopedic implants, dental applications, microfluidic systems; aviation/weather; electronics; energy; radioactive materials, etc.). Finally, Section 6 will identify and discuss any remaining areas where there is limited/existing knowledge in regards to research done on PEEK, followed by Section 7 providing a concise summary of the review. We conclude by outlining the research frontiers and directions of the future of PEEK, emphasising the need for innovative development in recyclability technologies, osseointegration-enabling surface treatment, and scaled-up manufacturing techniques. This review will provide researchers and engineers with a critical insight into the paradigm shift of PEEK in advanced materials science. It will propose potential innovations in polymer engineering and biomedical device development.

Table 1. *Global medical PEEK market valuation and growth projection from 2024–2033. Adapted from (Research, 2024).*

Year	Market Size (USD Billion)
2024	702.31
2025	753.58
2027	866.42
2029	995.87
2031	1,144.85
2033	1,324.12
CAGR: 7.30% (2025–2033)	

2. POLYMERIZATION TECHNIQUES AND CHEMICAL COMPOSITION

Polyetheretherketone (PEEK) is a polymer that possesses high-performance thermoplastic with superior thermal stability and mighty mechanical strength, which are owed in large measures to the unique aromatic backbone. PEEK molecular structure (Figure 1) consists of repeating monomer units that each contain three bridging aromatic rings and ether (-O-) and ketone (-CO-) linkages (Rahmitasari et al., 2017). Apart from conferring chemical resistance, this molecular arrangement also accounts for the superior performance of the material in extreme environments.

Researchers traditionally synthesise the compound PEEK through a nucleophilic attack on the benzene ring by phenoxide ions in a nucleophilic aromatic substitution reaction, which is depicted in Figure 1. Technology for this synthesis involves reacting 4,4'-difluoro benzophenone, and the disodium hydroquinone salt in a diphenyl sulfone solvent under the high heat of approximately 300°C (Kemmish, 2010; Ahmad et al., 2018). Synthesis occurs through several steps: (i) activation of the hydroquinone to give rise to disodium hydroquinone, (ii) attack by phenoxide ions on difluoro-benzophenone, which substitutes for the fluoride atoms, and propagation through sequential ether and ketone bond formation.

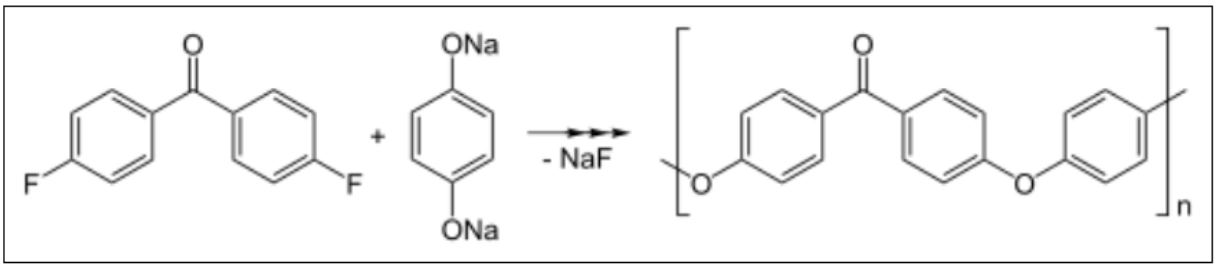


Figure 1. *Structural representations and Synthesis of PEEK via nucleophilic aromatic substitution reaction between 4,4'-difluorobenzophenone and hydroquinone disodium salt (Ahmad et al., 2018).*

Key considerations influence the yield and efficiency of the polymerisation reaction. The reaction temperature must be regulated very accurately to approximately 300°C if it is to polymerise without any heat degradation. Diphenylsulfone was also utilised as a solvent due to its high boiling point and compatible polarity. In addition, a strict 1:1 molar monomer ratio is critical in a bid to achieve polymers with high molecular weight and reproducible properties.

Table 2. *Classification of materials used for PEEK composites (Francis et al., 2022).*

Category	Materials
PEEK Composites	Carbon Based, Polymer Based, Ceramic Based, Metallic
Carbon Based	Carbon Fibres, Graphene, Carbon Nanotubes, Graphene Nanoplatelets
Polymer Based	Polytetrafluoroethylene (PTFE), Polyetherimide (PEI), UHMW Polyethylene (PE), Polysulfone (PSU), Polylactic Acid (PLA)
Ceramic Based	Hydroxyapatite (HA), Metallic Salts, Tricalcium Phosphate (TCP), Zirconia, Tantalum Pentoxide (Tantala), Alumina, Silicon Nitride, Mica
Metallic	Titanium, Silver, Magnesium, Zinc

After polymerisation, processing may occur for PEEK in the amorphous or semicrystalline state, based on the quenching conditions. Fast quenching generally produces an amorphous material with a crystallinity of ca. 12%, while slow and controlled quenching produces a semicrystalline material with crystallinity of as much as 48% (Jones et al., 1985; Bassett et al., 1988). In addition to conventional synthesis, electrophilic processes and other additive manufacturing (AM) methods are emerging for PEEK manufacture. AM especially benefits orthopaedic devices by enabling custom implants of intricate geometries specific to individual anatomy (Pulipaka et al., 2023; Jiang et al., 2024). Its ability to design directly from digital models is a critical aspect of personalised biomedical engineering. In this context, Figure 2 classifies the various materials used in PEEK composites (i) metallic, (ii) carbon-based, (iii) polymeric, and (iv) ceramic. Numerous composites are intrinsically fortified with distinct mechanical, physicochemical, and biological properties, highlighting their distinct contributions to the composite's performance. (Francis et al., 2022).

2.1. Synthesis via Nucleophilic Aromatic Substitution Polycondensation

Poly(ether ether ketone) (PEEK) is a semicrystalline thermoplastic characterised by alternating ether (−O−) and ketone (−C(= O)−) units along its backbone. It is well known for its exceptional thermal stability, mechanical strength, and chemical resistance (Talanikar et al., 2023). Due to these advantageous properties, PEEK is an indispensable material in demanding applications such as aerospace components, medical devices, and high-temperature industrial equipment. PEEK is typically synthesised via step-growth polymerisation through nucleophilic aromatic substitution (NAS), where bisphenol monomers react with activated dihalides such as 4,4'-difluorobenzophenone in an alkaline environment. In this mechanism, the hydroxyl groups on the aromatic rings nucleophilically attack the electrophilic carbonyl carbons, forming stable ether linkages with the concomitant elimination of halide byproducts.

Recent research has demonstrated the effectiveness of strained bicyclic monomers, such as 4,4'-(bicyclo[2.2.1]hept-5-en-2-yl)methylene bis(2-methoxyphenol) (BPA-NB), in producing PEEK derivatives with improved properties. The cross-link polymerization process for PEEK-NB (Figure 2) shows how strained monomers incorporate into the polymer backbone to enhance solubility and tune thermal transitions. Post-polymer modification via azide-alkene cycloaddition reactions enables the incorporation of functional moieties such as coumarilyl and anthracenyl groups (Talanikar et al., 2023).

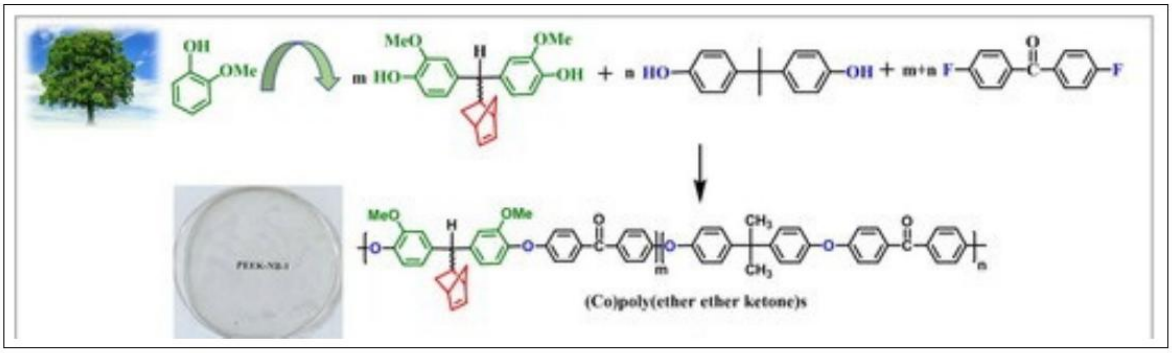


Figure 2. *Cross-link Polymerization of PEEK-NB (Talanikar et al., 2023).*

2.2. Cross-linking Techniques

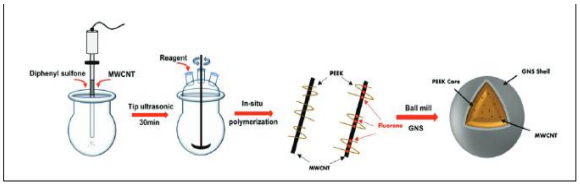
Cross-linking techniques significantly enhance poly(ether ether ketone) (PEEK) properties for advanced composites. The Friedel-Crafts alkylation reaction introduces C-C cross-links between phenyl groups via formaldehyde dimethyl acetal (FDA) and FeCl3 catalyst at 30°C, increasing residue mass by 7.5% at 900°C (Yamamoto et al., 2024). XPS analysis confirms cross-link formation through increased 284.6 eV C-C peaks. Table 3 presents the quantitative XPS analysis of bond formation in PEEK before and after cross-linking, showing the relative increase in C-C and C-O bonds that confirm successful cross-linking. When applied to carbon fibre-reinforced thermoplastics (CFRTPs), optimised hot-pressing at 370°C with 30 vol% fibers yields composites with 53.8% higher flexural strength than untreated PEEK, while maintaining 70% mass retention post-pyrolysis. The cross-linked PEEK matrix improves impregnation at the fibre interface, reducing void formation despite increased viscosity (Yamamoto et al., 2024).

Table 3. *XPS analysis of bond formation in PEEK before/after cross-linking (Yamamoto et al., 2024).*

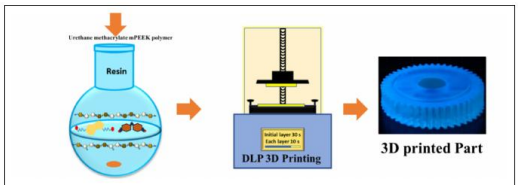
Bond Type	PEEK (rel. amount)	Cross-linked PEEK
C-C (284.6 eV)	3.11	5.66 (+82%)
C-O (286.2 eV)	0.31	0.84 (+171%)
C=O (531.5 eV)	1.00	1.00 (reference)

2.3. Additive Manufacturing of PEEK

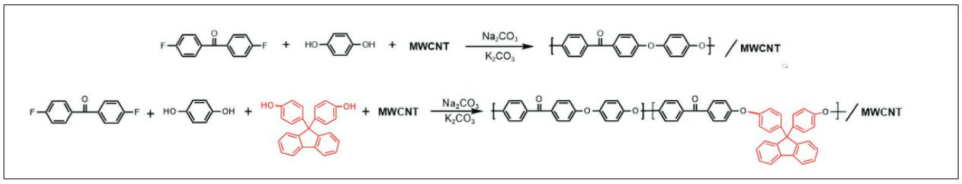
Advances in additive manufacturing (AM) processes in the last few years have revolutionized the processing of polyetheretherketone (PEEK) using three primary processes. Fused filament fabrication (FFF) is the most convenient process available to date but is also subject to advancements such as the use of hollow filaments with discontinuous reinforcement (Jiang et al., 2024) or ASA/PEEK filaments with 25% increased tensile strength (39 MPa) compared to neat polymer (Palacios-Ibáñez et al., 2022). A breakthrough in applying the less expensive photocurable PEEK resins to digital light processing (DLP) technology, depicted in Figure 3b, has achieved a 2.1 GPa Young's modulus (55% of bulk PEEK) using room-temperature-based processing (Pal et al., 2024). Selective laser sintering (SLS) based on powders obtained through thermally induced phase separation (TIPS) provides superior flowability (32° angle of repose) and 12% increased crystallinity with a 0.1 wt% addition of CNTs (Yamamoto et al., 2024). All these techniques allow accurate surface properties and inner architecture control in PEEK structures, which can be highly beneficial in biomedical applications in which surface modification (phosphonation/sulfonation) improves bioactivity and osseointegration (Dondani et al., 2023). The selection between these techniques depends on required resolution and mechanical properties as well as thermal stability, with each process bringing its distinct advantages to applications from load-bearing implants to complex tissue scaffolds.



(a) *In situ* grafting polymerization for PEEK/MWCNT nanocomposites showing covalent bonding between DB-functionalized MWCNTs and PEEK matrix. Adapted from (Jiang et al., 2023a).



(b) DLP workflow for PEEK: (i) resin formulation, (ii) UV patterning, (iii) post-curing. Reproduced from (Pal et al., 2024).



(c) Fluorene-functionalized PEEK (FD-PEEK) synthesis via tip-ultrasonic-assisted *in situ* polymerization, demonstrating enhanced MWCNT dispersion through π–π interactions. Adapted from (Jiang et al., 2023b).

Figure 3. *Advanced PEEK fabrication techniques:* ***a)*** *Covalent grafting methodology for nanocomposites;* ***b)*** *Digital Light Processing (DLP) additive manufacturing workflow;* ***c)*** *Fluorene-functionalized PEEK synthesis with ultrasonic dispersion.*

2.4. Nanocomposite Synthesis

Recent developments in PEEK nanocomposite processing utilise covalent and non-covalent methodologies to enhance nanofiller dispersion and interfacial bonding. Two complementary approaches have emerged for PEEK nanocomposite synthesis, as summarized in Figure 3.

The first approach, in situ grafting polymerization (Figure 3a), covalently bonds 4,4'-difluorobenzophenone (DB) onto MWCNTs. This process leads to a 28% increase in tensile strength with 1 wt% MWCNT loading and an increase in glass transition temperature to 153°C due to chain mobility restriction (Jiang et al., 2023a).

The second approach utilizes fluorene-functionalized PEEK (FD-PEEK) with π-π stacking (Figure 3c) to non-covalently anchor MWCNTs. This method achieves a thermal conductivity of 2.41 W m^{-1} K^{-1} (868% increase compared to neat PEEK) while preserving tensile strength (104.3 MPa) at 10 wt% filler load (Jiang et al., 2023b).

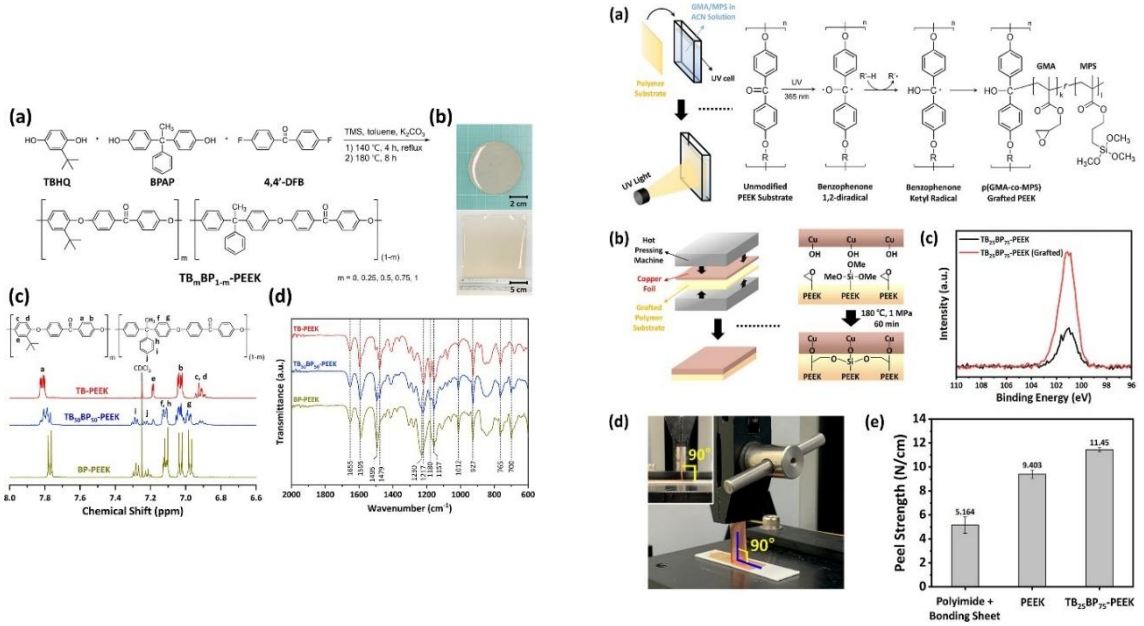
The covalent strategy outperforms in mechanical reinforcement and thermal stability (5% weight loss upon heating to 500°C), while fluorene modification offers higher filler load with good processability. Both procedures overcome interfacial resistance but in different ways: the grafting polymerization process involves alkaline metal-based catalysis with stoichiometric control, whereas fluorene modification employs sonication processing to enable scale-up for nanocomposite production.

2.5. Surface Modification and Film Synthesis

New advancements in PEEK allow for significant application in high-frequency electronics through two complementary methods: dielectric property modification and enhancement of interfacial adhesion. Low-dielectric-constant (low-k) PEEK copolymers, fabricated through nucleophilic aromatic substitution of tert-butylhydroquinone (TBHQ) and 4,4'-(α-methylbenzylidene)bisphenol (BPAP), achieve outstanding dielectric constants of 2.49 with losses of 5.84×10^{-3} at 28 GHz (Jang & Seo, 2023). Figure 4a(a) demonstrates that incorporation of bulky side chains disrupts crystalline packing, increasing free volume while maintaining thermal stability (Td > 458°C) and hydrophobicity (<0.5% water absorption).

Parallel developments in metal bonding utilise UV-activated surface grafting of 3-(trimethoxysilyl)propyl methacrylate (MPS) and glycidyl methacrylate (GMA), exploiting PEEK's intrinsic benzophenone moieties as photoinitiators. This yields peel strengths of 11.45 N/cm (Figure 4b(b)), a 2.2-fold improvement over polyimide laminates (Jang & Seo, 2023).

Comparative analysis reveals trade-offs: while TBxBPy-PEEK copolymers exhibit increased elastic modulus (1.64 GPa vs. 0.43 GPa for neat PEEK), they maintain low dielectric constants (k = 2.49) with minimal post-grafting increase (Δk ≈ +0.14). The optimal TB25BP75-PEEK composition shows balanced properties with 0.5% water uptake.



(a) Synthesis route for low-*k* TB_xBP_y-PEEK copolymers, showing molecular design with bulky tert-butyl (blue) and phenyl (red) groups to disrupt crystallinity.

(b) UV-initiated surface grafting mechanism for copper adhesion enhancement, utilizing PEEK's benzophenone moieties to anchor methacrylate polymers. Peel strength comparison shows 2.2× improvement over polyimide.

Figure 4. *Combined figure showing* ***a)*** *low-k PEEK copolymer synthesis and* ***b)*** *UV-grafting for enhanced adhesion. Adapted from (Jang & Seo, 2023).*

2.6. Extrusion and Injection Moulding

Extrusion is the technique of forming material by means of forced die shaping. Using a die, this method extrudes poly-ether-ether-ketone (PEEK) to create continuous forms, including sheets or filaments. Usually used in conjunction with additive manufacturing processes like Fused Deposition Modeling (FDM), this method is quite helpful for building components with consistent cross-sections (S. Li et al., 2022). In contrast, injection moulding, elevating the polymer's temperature beyond its melting point and injecting it into a mould, helps this approach to be used extensively for manufacturing PEEK components. Small-scale manufacturing finds this ideal since it helps to fabricate complex geometries and exact forms. This work focuses on composites made using the injection moulding technique, allowing the construction of standard test bars from the obtained grains (He et al., 2024; S. Li et al., 2022). (Table 4) represents the principal differences between injection moulding and extrusion.

Table 4. *Comparison of Processing Techniques for PEEK.*

Parameter	Injection Molding	Extrusion
Process Type	Batch Process	Continuous Process
Applications	Complex shapes	Continuous profiles
Advantages	High precision, scalability	Versatility, material blending
Disadvantages	High tooling costs	Limited shape complexity

Table 5. *Advanced Techniques for PEEK Processing and Applications.*

Technique	Advantages	Applications	References
Step-Growth Polymerization	High molecular weight, simple process, cost-effective	Aerospace, medical devices, industrial components	(Talanikar et al., 2023)
Cross-Linking Techniques	Enhanced thermal/mechanical properties, high gel content	High-temperature applications, CFRTP composites	(Yamamoto et al., 2024)
Additive Manufacturing	Complex geometries, rapid prototyping, customization	Space, 3D printing, medical devices	(Pulipaka et al., 2023; Jiang et al., 2024; Pal et al., 2024)
Nanocomposite Synthesis	Improved properties, better nanofiller dispersion	Aerospace, automotive, biomedical	(Jiang et al., 2023a; 2023b)
Surface Modification	Low dielectric constant, high copper adhesion	Electronics, high-frequency comms	(Jang & Seo, 2023)

3. PHYSICAL AND MECHANICAL PROPERTIES OF PEEK COMPOSITES

Polyetheretherketone (PEEK) is a high-performance engineering polymer with excellent mechanical, thermal, and tribological properties. This section discusses the thermal stability, mechanical strength, tribological properties, and influence of crystallinity on the performance of PEEK composites.

3.1. Analytical Methods for PEEK

Characterisation of material properties often commences with Thermogravimetric Analysis (TGA), which examines PEEK's thermal stability and decomposition profile. PEEK composites with Halloysite nanotube reinforcement have been reported to possess greater thermal stability compared to neat PEEK (Siraj et al., 2023). Differential Scanning Calorimetry (DSC) examines PEEK's crystallisation behaviour and thermal transitions. Crystallinity can be quantitatively determined using DSC, ranging from 26.53% to 66.57%, depending on material composition and process conditions (El Magri et al., 2021; Lu et al., 2024). Dynamic Mechanical Analysis (DMA) measures the viscoelastic behaviour of PEEK composite materials, which shows that nano-silica reinforced PEEK composites can show an increase in dynamic modulus by 40% at high temperatures (Lai et al., 2007).

Scanning Electron Microscopy (SEM) examines PEEK composite materials' morphological surface features and failure mechanisms. As an illustration, analysis by SEM of PEEK-30CF composite shows good fibre-

matrix adhesion with negligible fibre pull-out (Maslavi et al., 2024). Transmission Electron Microscopy (TEM) analyses nanoparticle dispersion in PEEK matrices. An investigation by TEM on PEEK composite materials reinforced using nano-silica proves uniform particle dispersion and high interfacial interactions (Lai et al., 2007). In addition, X-ray diffraction (XRD) is used to examine PEEK's crystallinity and crystal structure, including that of its composites, indicating that CF/PEEK composite possesses a 35.5% increase in crystallinity over neat PEEK (Sariyev et al., 2023). Mechanical test techniques play an important role in measuring PEEK composite performance. Tensile Testing tests tensile strength, modulus, and elongation, such that, for example, tensile strength is 796.1 MPa, Young's modulus is 42.3 GPa for CF/PEEK composite (Yamamoto et al., 2024). Impact Testing measures toughness and impact resistance for PEEK-30CF composite, which shows improved impact strength over neat PEEK (Padhan et al., 2022). Finally, Nanoindentation tests test hardness and Young's modulus, indicating that PEEK/graphite laminates also have higher hardness and Young's modulus than neat PEEK (Sariyev et al., 2023). These analytical methods complement one another to address an exhaustive knowledge synthesis for PEEK and its composite materials, enabling their development towards practical application.

3.2. Thermal Stability and Melting Behavior

PEEK is known for its high degree of thermal stability, with approximately 343°C melting point, making it suitable for heat degradation-resisting applications (F. Li, Hu, Hou, Hu, & Jiang, 2018). Crystallinity affects thermal properties, with higher crystallinity often resulting in higher thermal stability (Regis et al., 2017). PEEK is rendered more stable by annealing, resulting in higher crystallinity in the material structure (Rinaldi et al., 2021). PEEK melts in response to thermal transitions while experiencing negligible damage, with standard melting points between 341°C to 350°C (Doumeng et al., 2019). Filers or additives such as carbon fibers may alter thermal properties in a composite material (Wang, 2023). PEEK's thermal properties may be determined using Differential Scanning Calorimetry or DSC, yielding data on melting point, crystallization point, as well as thermal transitions (Pulipaka et al., 2023).

Table 6. Thermal Properties of Polyetheretherketone (PEEK).

Property	Value	Notes	Reference
Glass Transition Temperature (Tg)	143°C	Transition from brittle to rubbery state	(Chai et al., 2022)
Melting Temperature (Tm)	337°C to 343°C	337.7°C for unirradiated samples	(Y. Yao et al., 2022)
Crystallization Temperature (Tc)	~290°C	Temperature for crystalline solid formation	(X. Yao et al., 2022)
Enthalpy of Fusion (ΔH)	~130 J g−1	Derived from DSC melting peak area	(Blundell & Osborn, 1983)
Effect of Irradiation	Changes in Tg and Tm	Observed with increasing absorbed doses	(Rashed et al., 2024)

3.3. Comparison with Other Materials

Building upon the thermal properties of PEEK, it is essential to compare these characteristics with other high-performance materials to evaluate PEEK's relative advantages. PEEK is often compared with other high-performance polymers such as poly(aryl ether ketone) (PAEK) and poly(phenylene sulfide) (PPS). CF/PEEK composites exhibit superior mechanical properties compared to CF/PAEK and CF/PPS composites, with a tensile strength of 796.1 MPa and a Young's modulus of 42.3 GPa (Ramaswamy et al., 2023; Lu et al., 2024). These mechanical enhancements make CF/PEEK an ideal material for demanding applications.

Compared to thermosetting composites, thermoplastic composites based on PEEK have numerous advantages, such as quicker processability times, increased impact strength, and ease of recyclability. For example, compared to epoxy-based composites, CF/PEEK composite material possesses enhanced interlaminar shear strength (ILSS) and mode I fracture toughness (Ramaswamy et al., 2023; Yamamoto et al., 2024). These advantages render PEEK composites more versatile for use in sectors where performance, longevity, and efficient processability are necessary.

Additionally, in contrast to metal alloys, PEEK composites have been used more often in recent times as lightweight materials. As an illustration, CF/PEEK composite has been found to have better strength-to-weight ratios compared to aluminum and steel alloys (Lu et al., 2024; Osti de Moraes et al., 2023). This makes them highly suitable for use in areas where light-weighting is desirable as much as withstanding harsh environmental conditions. While PEEK's thermal properties are impressive, its true engineering value emerges when compared to alternative materials, as shown in Table 7.

Table 7. *Material Comparison and Key Findings.*

Material	Comparison	Key Findings	Citation
PEEK vs. PI	Flexural properties	PEEK has higher flexural strength and modulus (0.4 GPa vs. 0.2 GPa)	(Kosmachev et al., 2021)
PEEK vs. Epoxy	Strain energy	PEEK exhibits better strain energy storage and plasticity	(Hernandez, Mills, & Yazdani Nezhad, 2018)
PEEK vs. PMMA	Flexural & hardness	PEEK has higher flexural strength (183 MPa vs. 84 MPa) and hardness (24 VHN vs. 19.4 VHN)	(Mutneja et al., 2021)
CF/PEEK vs. CF/Epoxy	Interfacial shear	CF/PEEK has higher interfacial shear strength and mode I fracture toughness	(Ramaswamy et al., 2023)
PEEK vs. Titanium	Implant fatigue	Titanium has higher fatigue strength and safety factors for implants	(A et al., 2023)
Hybrid PEEK/PEI vs. Single PEEK	Delamination	Hybrid composites show improved strain to failure and delamination resistance	(Toro et al., 2024)

3.4. Mechanical Strength

3.4.1. Tensile and Flexural Strength and Impact Resistance

PEEK exhibits a tensile strength of around 95 MPa (Table 8), which can be enhanced by incorporating fillers such as modified silicon dioxide (m-SiO2) and graphene oxide (m-GO). For instance, PEEK composites with 5.0 wt% m-SiO2 and 1.0 wt% m-GO demonstrate a tensile strength increase to 99.43 MPa (Rashed et al., 2024); additionally, the tensile strength of PEEK can be improved by reinforcement with carbon fibers. PEEK composites containing 20 wt% short carbon fibers (SCFs) have demonstrated considerable enhancements in tensile strength, achieving up to 209 MPa, contingent upon fiber length and processing technique (Wang, 2023). Neat PEEK exhibits a flexural strength of approximately 140.62 MPa, which can be enhanced by incorporating fillers. PEEK composites containing 5.0 wt% m-SiO2 and 1.0 wt% m-GO have a flexural strength of 146.06 MPa (Rashed et al., 2024). The flexural strength of PEEK composites can be further augmented using carbon fiber reinforcements. Composites containing 30 wt% carbon fibers have exhibited flexural strengths of 254 MPa (Jeon et al., 2021). The impact resistance of PEEK can be preserved or enhanced by incorporating specific fillers. Integrating multi-walled carbon nanotubes (MWCNTs) into PEEK has demonstrated the ability to preserve or marginally enhance impact strength while augmenting other mechanical properties, such as flexural strength (Guney Yilmaz et al., 2024). Furthermore, PEEK composites augmented with graphene nanoplatelets (GNPs) exhibit remarkable mechanical properties, including improved impact resistance, attributable to greater interfacial contacts between the matrix and the fillers (Rashed et al., 2024).

Impact resistance is another critical mechanical property of PEEK composites. The addition of reinforcing fibers increases the tensile and flexural strength and enhances the material's ability to withstand sudden impacts, making it suitable for demanding applications (Doumeng et al., 2019).

3.5. Tribological Properties

PEEK demonstrates minimal friction and wear rates, which may be enhanced by using fillers such as PTFE, carbon fibers (Figure 5), and nanoparticles (Mir & Charoo, 2019). The wear resistance of PTFE/PEEK composites is significantly improved, shown by reduced wear rates in many experiments. This enhancement is ascribed to developing a stable transfer coating during friction, which safeguards the surface against abrasion (Qi et al., 2023; L. Yuan et al., 2024). Carbon fibres and nanoparticles such as SiC, SiO_2 and ZrO_2 are often used to improve the tribological characteristics of PEEK (Mir & Charoo, 2019). The integration of carbon fibers markedly enhances wear resistance and decreases the coefficient of friction, particularly under elevated load conditions (Lin et al., 2022). Nano-sized fillers like SiO2 and ZrO2 facilitate the development of a homogeneous transfer coating, reducing wear and friction. PEEK and its composites exhibit varying performance in dry and lubricated environments. Although water lubrication may diminish the hardness of PEEK, it may concurrently facilitate wear owing to decreased surface hardness.

The inclusion of water or other lubricants may modify wear processes, with some research suggesting that lubrication may not consistently enhance PEEK's performance (Mir & Charoo, 2019). External factors,

including temperature and humidity, influence the tribological performance of PEEK. Elevated temperatures may enhance crystallinity, influencing wear characteristics (Doumeng et al., 2019). Recent research has investigated recycled carbon fibers in PEEK composites, revealing that these materials may attain superior tribological performance akin to that of virgin fibers (Lin et al., 2022).

Table 8. *Properties of PEEK Composites (de Leon et al., 2021).*

Property	**Value (Unit)**
Specific Gravity	1.30–1.37
Molding Shrinkage (Flow 23°C)	0.0010–0.016 m/m
Water Absorption (23°C, 24 h)	0.049–0.21%
Tensile Modulus (23°C)	2612–4770 MPa
Tensile Strength (Yield, 23°C)	72–111 MPa
Tensile Strength (Break, 23°C)	70–99 MPa
Flexural Modulus (23°C)	2909–4432 MPa
Flexural Strength (23°C)	116–155 MPa
Coefficient of Friction	0.080–0.53
Notched Izod Impact (23°C)	41–101 J/m
Unnotched Izod Impact (23°C)	411–2189 J/m
Deflection Temp. (0.5 MPa, unannealed)	155–300°C
Deflection Temp. (1.8 MPa, unannealed)	143–178°C
Volume Resistivity (23°C)	1.9–2.0 ×1016 Ω cm
Dielectric Strength (23°C)	14–20 MV/m
Dielectric Constant (23°C)	2.80–3.61
Dissipation Factor (23°C)	0–0.013

PEEK and its composites exhibit varying performance in dry and lubricated environments. Although water lubrication may diminish the hardness of PEEK, it may concurrently facilitate wear owing to decreased surface hardness (Mir & Charoo, 2019). External factors, including temperature and humidity, influence the tribological performance of PEEK. Elevated temperatures may enhance crystallinity, influencing wear characteristics (Doumeng et al., 2019). Recent research has investigated recycled carbon fibers in PEEK composites, revealing that these materials may attain superior tribological performance akin to that of virgin fibers (Lin et al., 2022).

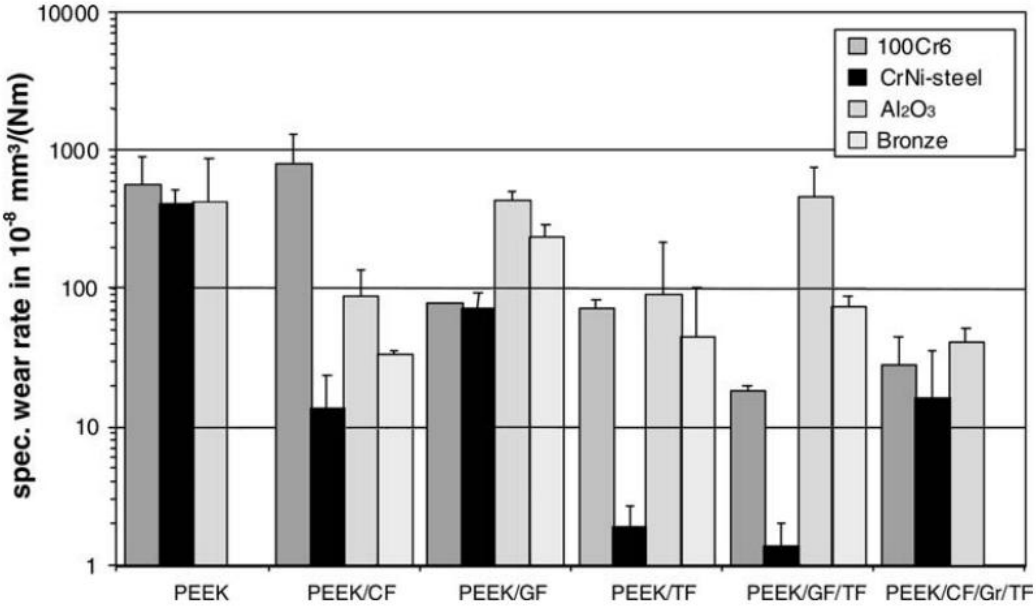


Figure 5. *Specific steady state wear rates of various PEEK compounds sliding against different counterparts under dry conditions (Jacobs et al., 2005).*

4. CHEMICAL AND ENVIRONMENTAL RESISTANCE

Polyetheretherketone (PEEK) is known for its superior chemical resistance, paving the way for use in aggressive environments. In its detailed chemical compatibility guide, Atlas Fibre states that PEEK shows high resistance to most chemicals, including acids, bases, halogens, and solvents. It, nevertheless, shows weaknesses against sulfuric acid at more than 10%, methylene chloride, dimethyl sulfoxide (DMSO), and tetrahydrofuran (THF) (Fibre, 2024). These findings are vital for choosing the relevant materials for corrosive agents.

Based on the chemical resistance, PEEK is further characterized by high resistance to a range of solvents, for example, organic solvents chloroform, acetone, and dimethylformamide (DMF), thus expanding its use in demanding situations even further. For example, due to solvent resistance, PEEK membranes can withstand 6 hours of immersion in chloroform without degradation of morphology over commercial PVDF and PPSU membranes (Yang et al., 2023).

In addition, nano-indentation experiments show that PEEK's surface hardness and elastic modulus are reduced step by step with exposure to acetone. However, its overall structural integrity remains intact (Iqbal et al., 2018). This is a testament that even with some difficulties caused by certain chemicals, PEEK is a durable material under different circumstances and is applicable in different industrial applications.

With the improved solvent resistance being witnessed, the resistance of PEEK can also be enhanced by structural modification. For instance, ternary plastic PEEK/PEI/PES alloys have been formulated for enhanced resistance to concentrated H2SO4, acetone, chloroform, and DMF depending on composition ratios (Chen,

2014). Surface modification of PEEK with hydrophilic polymers such as poly(acrylic acid) (PAA) increases anti-fouling properties while maintaining solvent resistance (Yang et al., 2023).

PEEK is highly resistant to both acidic and basic chemicals. In particular, PEEK membranes show over 90% rejection rates under acidic environments (for example, using 2 M HCl) with minimal degradation (Lin et al., 2023). In everyday environments (for example, using 2 M NaOH), PEEK maintains its structural integrity after going through high temperatures (Lin et al., 2023). Therefore, PEEK is a good material under aggressive chemical filtering and separation environments. This stability is due to the resonance-stabilized macromolecular structure of PEEK, which facilitates the delocalization of the electrons within the polymer chain. Thus, extremely unreactive (Kurtz, 2012). The stability of PEEK is also supported by resistance against both polar and nonpolar lubricant base oils, with minimal degradation compared to other high-performance polymers, e.g., polyphthalamide (PPA) (Martins Ferreira et al., 2024). The resistance against oxidation is another important property of PEEK, and the aromatic, semi-crystalline backbone structure endows it with superior thermal and oxidative stability even at the high temperatures of the oilfield environment (Yuan et al., 2013).

PEEK has high stability under radiative conditions, with gamma radiation to a dose of 20 MGy resulting in surface oxidation but little change to mechanical properties (Chai et al., 2022). The stability of PEEK makes it well adapted for use in nuclear and space environments where radiation is unavoidable.

In addition, PEEK's resistance against hydrolysis is crucial for application under aqueous environments since it withstands seawater and freshwater hydrolytic degradation, even at high hydrostatic stress (Liu et al., 2018). When subjected to boiling water, PEEK composites further prove the material's durability under extreme environmental conditions.

Table 9. *Summary of Chemical and Environmental Resistance of PEEK.*

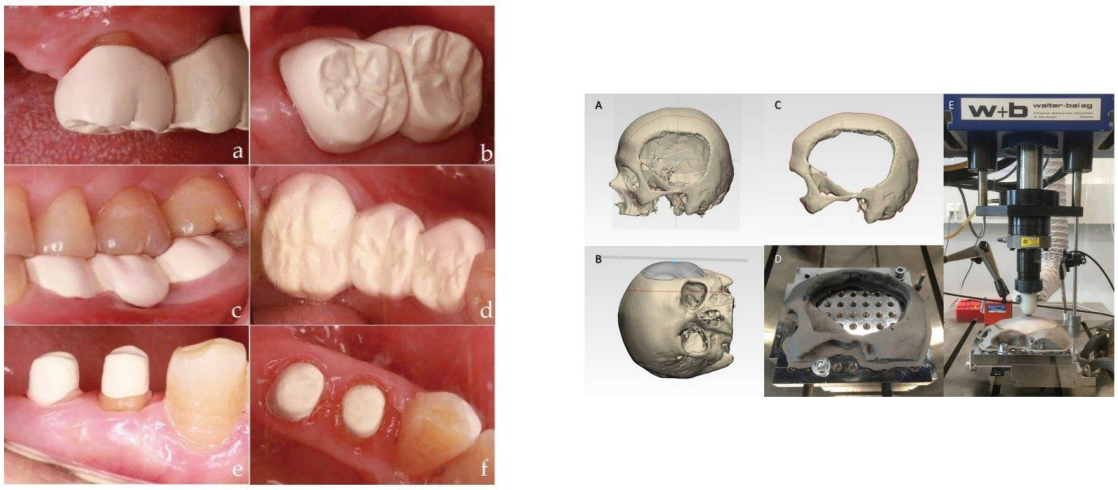
Property	Key Findings	Citation
Solvent Resistance	Resists chloroform, acetone, DMF, and other organic solvents; enhanced by PAA grafting	(Yang et al., 2023; Chen et al., 2022)
Acid and Base Resistance	Resists 2 M HCl and NaOH; minimal degradation in polar and non-polar oils	(Lin et al., 2022; Martins Ferreira et al., 2024)
Oxidation Resistance	Stable under gamma irradiation; resists oxidation in high-temperature environments	(Kurtz, 2012; Yuan et al., 2013)
Hydrolysis Resistance	Resists hydrolytic degradation in seawater and boiling water; Fickian diffusion	(Boinard, Pethrick, & MacFarlane, 2001)

5. RECENT ADVANCES IN PEEK RESEARCH AND ITS APPLICATIONS ACROSS INDUSTRIES

The versatility of polyetheretherketone (PEEK) has led to its widespread adoption across multiple industries, where it is utilised in various forms, including 3D-printed components, composites, and traditionally manufactured parts.

5.1. Medical and Biomedical Applications

PEEK is widely recognised for its biocompatibility and is extensively used in medical applications, including implants, surgical instruments, and custom prosthetics. Its elastic modulus closely matches that of human bone, minimizing stress shielding effects and enhancing its suitability for orthopedic and dental applications (Yang et al., 2025). In dental implantology, PEEK has emerged as a promising alternative to traditional materials. A comprehensive critical review by Alqutaibi et al. (2025) examined the applications, modifications, and manufacturing of PEEK in dental implantology, highlighting its advantages in terms of mechanical compatibility with bone and the potential for surface modifications to enhance osseointegration. Similarly, in dentistry (Figure 6a), it serves as a material for implants, abutments, and frameworks due to its biocompatibility and resistance to degradation (Wang et al., 2022). Figure 6 shows clinical and biomechanical applications of PEEK in dental and cranial surgery.



(a) Clinical photographs of PEEK prostheses (B. Wang et al., 2022).

(b) Biomechanical characterisation steps for PEEK implants: skull model design, implant orientation, template design, and 3D printing (Sharma et al., 2021).

Figure 6. *PEEK in dental and cranial biomedical applications:* ***a)*** *clinical use in prosthetics (Wang et al., 2022),* ***b)*** *biomechanical implant testing and preparation (Sharma et al., 2021).*

The challenge of PEEK's biological inertness has been a major focus of recent research. Sun et al. (2025) provided a comprehensive overview of multifaceted strategies to optimize PEEK bioactivity, including surface modifications, composite formulations, and immunomodulatory approaches for interdisciplinary clinical applications. Zhang et al. (2025) demonstrated that hierarchical micro-/nanotopographical cues on PEEK implants can significantly enhance both osteogenesis and angiogenesis, addressing the critical need for

vascularization in bone healing. Their work showed that precisely engineered surface topographies can create a pro-healing microenvironment that favors osseointegration over fibrous encapsulation.

Recent studies on the tribological properties and biocompatibility of PEEK and its composites by SK (S K, 2025) have confirmed their suitability for load-bearing biomedical applications, demonstrating favorable wear resistance and cell compatibility under physiological conditions.

Table 10. *Biomedical Applications of Modified PEEK in Clinical Practice (Yang et al., 2025).*

Application Category	Clinical Uses and Functional Benefits
Bone Regeneration	Osteomyelitis treatment; Osteoporotic defect repair; Osteosarcoma reconstruction
Dental Solutions	Dental implants; Periodontal scaffolds
Cardiovascular Devices	Heart valve components; Vascular stents
Ophthalmic Implants	Intraocular lenses; Orbital implants
Therapeutic Delivery	Antibiotic-eluting devices; Chemotherapy carriers
Advanced Manufacturing	3D-printed implants: <100 µm resolution; Bioelectronic sensors: 12+ month stability

Similarly, in dentistry (Figure 6a), it serves as a material for implants, abutments, and frameworks due to its biocompatibility and resistance to degradation (Wang et al., 2022).

5.2. PEEK in Microfluidic Chip Applications

Microfluidics facilitates accurate microscale fluid manipulation, moving most laboratory processes to a chip. This ability to handle small amounts of fluids with precision is useful in chemistry, biology, and medicine. Microfluid flow in channels usually leads to laminar flow with low Reynolds numbers and stable fluid flow. Microchannels are designed with 2D and 3D computer-aided design software for particular purposes and simulations that predict fluid flow under diverse circumstances (Song et al., 2018).

Material selection and fabrication processes have significant effects on microfluidic performance. Some materials, including PDMS, glass, and, in particular, polyether ether ketone (PEEK), are favoured because they have high thermal, mechanical stability, and chemical resistance. PEEK is especially promising for organ-on-a-chip (OOC) use (Figure 7a), allowing for the development of multi-layer systems with high-performance bonding using techniques such as plasma-assisted hermetic sealing. This reduces risks of blockage and contamination (Awaja et al., 2018). Furthermore, well-bonded PEEK devices retain dimensional integrity, which means microchannel structures function as designed in applications in medicine, improving opportunities for personalized medicine (Awaja et al., 2018).

The biocompatibility and strength of PEEK render it ideally suited for point-of-care diagnostics, such as applications in microfluidic systems. PEEK's application in microfluidic systems for capillary electrophoresis (CE) devices improves performance and reliability in such diagnostic tools. Its thermal tolerance to

temperatures over 200°C means it can perform in harsh conditions where other materials cannot. In addition, 3D-printed PEEK can have customized microfluidic reactors designed, which is crucial in most biomedical uses where individualized fluid channels are needed.

(a) PEEK-based organ-on-a-chip device with microfluidic channels and reservoirs (Awaja et al., 2018).

(b) Structure dimensions of a PEEK microreactor (Menzel et al., 2020).

(c) PEEK chip with three CCD electrode pairs showing plasma-assisted thermal bonding.

Figure 7. *PEEK-based microfluidic device applications:* ***a)*** *organ-on-a-chip layout (Awaja et al., 2018),* ***b)*** *microreactor structure (Menzel et al., 2020),* ***c)*** *PEEK chip with CCD electrode pairs showing plasma-assisted thermal bonding.*

In microchip applications like CE chips, PEEK has effectively bonded films with different crystallization levels. This aspect is important in sealing microchannels and making the device leak-free. Thermal bonding, where plasma treatment is applied, has been used to seal PEEK-based microchips with stable, high-performance devices for quick diagnostic tests. The use of PEEK in diagnostic systems, presented in Figure 7c, illustrates the capability of such devices for reliable, real-time results for use in clinics (Kosloh, Sackmann, & Schomburg, 2007).

In other areas, Polyether ether ketone (PEEK) has also revolutionised flow chemistry by enabling customised and affordable microfluidic reactors. Such reactors enhance mixing efficiency and accommodate multistep reactions to suit complex chemical transformations. In particular, creating an end-to-end 3D-printed flow system, including such components as separators and back-pressure regulators, combats accessibility in commercial equipment. Also, experiments for targeted reactions, such as fluorination of ribose derivatives, realise high product yields, especially for synthesising nucleoside anti-cancer medicines. Successful

miniaturisation to less than 500 µm straight channels (Figure 7b) highlights further opportunity for more efficient flow systems, encouraging wider practice in this emerging area and providing practical alternatives to conventional practice in flow chemistry processes (Menzel et al., 2020).

Overall, the use of PEEK in organ-on-a-chip devices not only enhances the performance and reliability of the devices but also opens up new avenues in advanced research in microfluidics and in the biomedical field.

5.3. Marine and Aerospace Applications

PEEK demonstrates favorable tribological characteristics in bearings and gears, exhibiting reduced friction and wear rates when in contact with corrosion-resistant 316 steel under seawater lubrication. This results from the accumulation of calcium carbonate and magnesium hydroxide on the counterface, which amplifies the lubricating properties of seawater. Increased friction and wear rates are noted when in contact with GCr15 steel (Chen et al., 2011). Researchers have developed PEEK coatings embedded with inorganic fullerene-like tungsten disulfide (IF-WS2) nanoparticles to enhance their tribological properties in marine environments, leading to diminished friction and wear rates, thereby rendering them appropriate for high-temperature and high-load marine applications. Underwater equipment often operates in water-lubricated environments, where the tribological performance of PEEK is crucial. Studies demonstrate that neat PEEK and its composites, such as 30% carbon fibre-reinforced PEEK, have decreased coefficients of friction and wear rates in water-lubricated conditions compared to dry sliding scenarios. Carbon fibres markedly enhance the wear resistance of PEEK, especially at elevated loads and sliding velocities (Unal & Mimaroglu, 2006). The selection of counterface material is essential for the tribological efficacy of PEEK in aquatic applications. PEEK composites reinforced with carbon fibers demonstrate decreased wear rates in water-lubricated environments when in contact with alumina (Figure 8). This arises from the enhanced wear resistance of carbon fibers in aquatic settings (Jacobs et al., 2005).

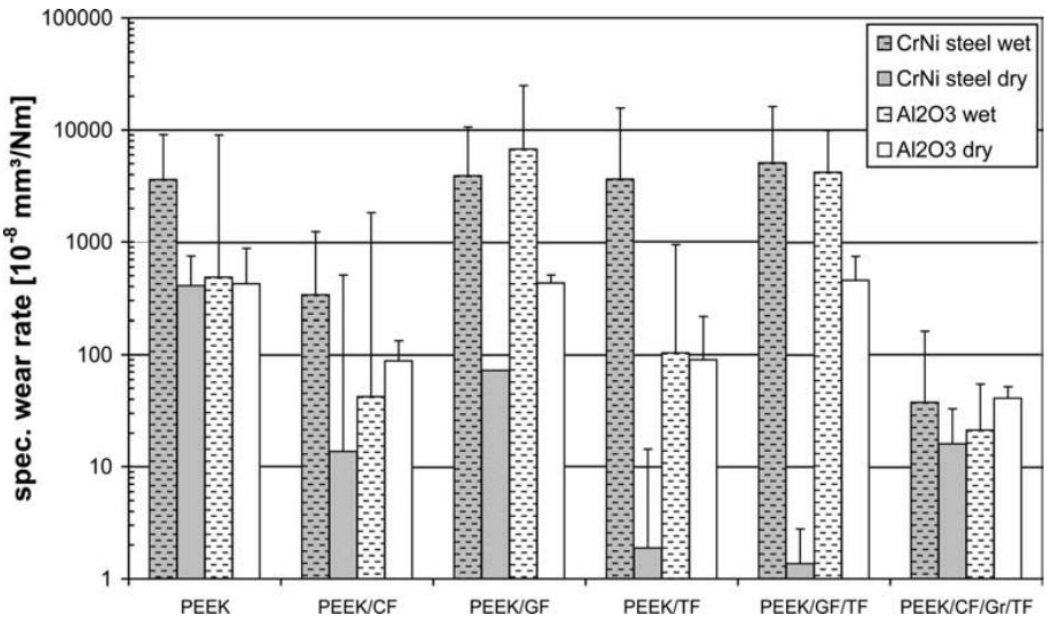


Figure 8. *Steady state wear rates of various PEEK compounds after sliding against CrNi steel and against alumina, respectively (Jacobs et al., 2005).*

In modern aerospace engineering, composite materials have been commonly used to enhance aircraft performance. For example, over 53% of the Airbus A350 is built with composite materials to reduce weight and enhance fuel usage. Among all composite materials, PEEK (Polyether Ether Ketone) stands out due to unique properties that qualify it as a suitable material for high-performance aerospace operations (Asyraf et al., 2022).

Figure 9 shows the composite material application in Airbus aircraft, reflecting on how composite materials are efficient in aerospace development. PEEK material properties equate to aerospace component development for resistance to high temperatures and harsh chemical exposures. PEEK materials have been shown through studies to have been utilized in fuel systems and interior cabin areas due to their thermal and chemical resistance (Precision, 2021). Composite PEEK solutions have been applied more in aerospace for structure-related components with weight savings without losing strength requirements (plc, 2025).

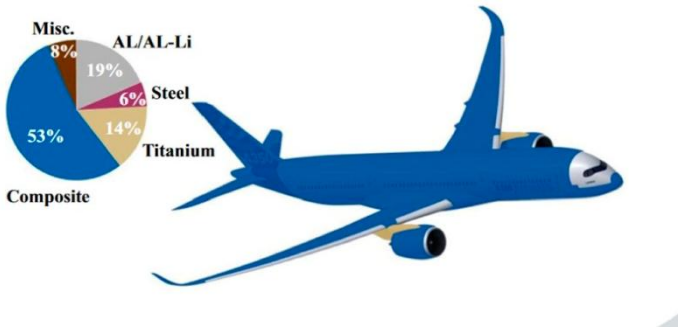


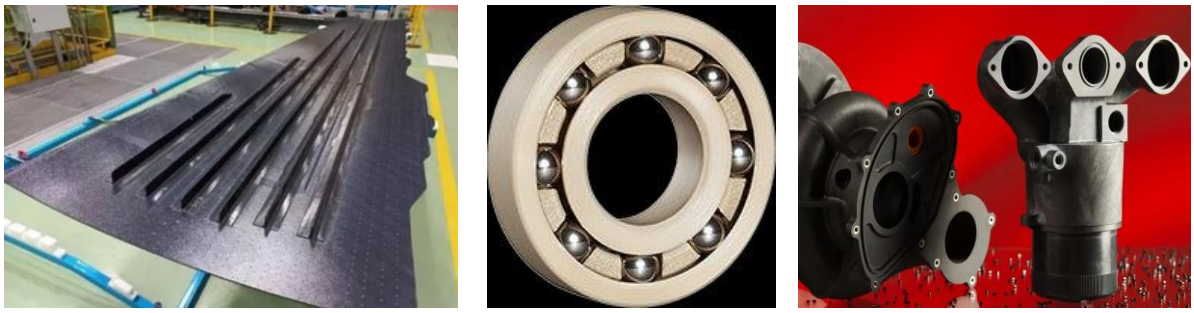
Figure 9. *Application of composite materials in Airbus aircraft, demonstrating their role in advancing aerospace technology (Asyraf et al., 2022).*

Based on the unique properties of PEEK, in the Clean Sky 2 OUTCOME project, Figure 10a showed that PEEK thermoplastic could be used effectively in an aircraft wing box cover for an Airbus C-295, signalling its application in aerospace technology advancement (project, 2021). PEEK composite material has also been reported to decrease the weight of aircraft structures to improve fuel efficiency and operational performance (AZoM, 2021).

The capability of PEEK is still further highlighted by its application in precision mechanical components, including ball bearings and gears (Figure 10b), which in high-performance settings play indispensable roles. Recent studies have found PEEK bearings combined with stainless steel balls to be more efficient and longer-lasting in various mechanical processes (EMC, 2021). PEEK material development has also led to even more cost-effectiveness and efficiency in the production of crucial aerospace components (Hub, 2021).

Fusible-core injection moulding has been applied to produce an aircraft fuel housing from Victrex PEEK material with 30% cost savings and 50% weight reduction over metallic constructions (Figure 10c). Egmond Plastic consolidated parts, integrating components like the bearing into the housing. This process facilitates

complex geometries and shortens lead times by 50%, with more than 30% cost savings over aluminium components, while it fulfils stringent requirements in engineering. This process has been proven to have an application in Eurofighter Typhoon's fuel pump, which has been in service for more than 20 years (Hutton, 2015).



(a) Clean Sky 2's PEEK Airframe (project, 2021).

(b) PEEK bearing featuring stainless steel balls (of Composite Science, 2020).

(c) PEEK fuel housing (Hutton, 2015).

Figure 10. *Examples of PEEK utilization in aerospace, showcasing its application in both high-performance materials for aircraft and precision components like bearings:* ***a)*** *Clean Sky 2's PEEK Airframe (project, 2021),* ***b)*** *PEEK bearing featuring stainless steel balls,* ***c)*** *PEEK fuel housing (Hutton, 2015).*

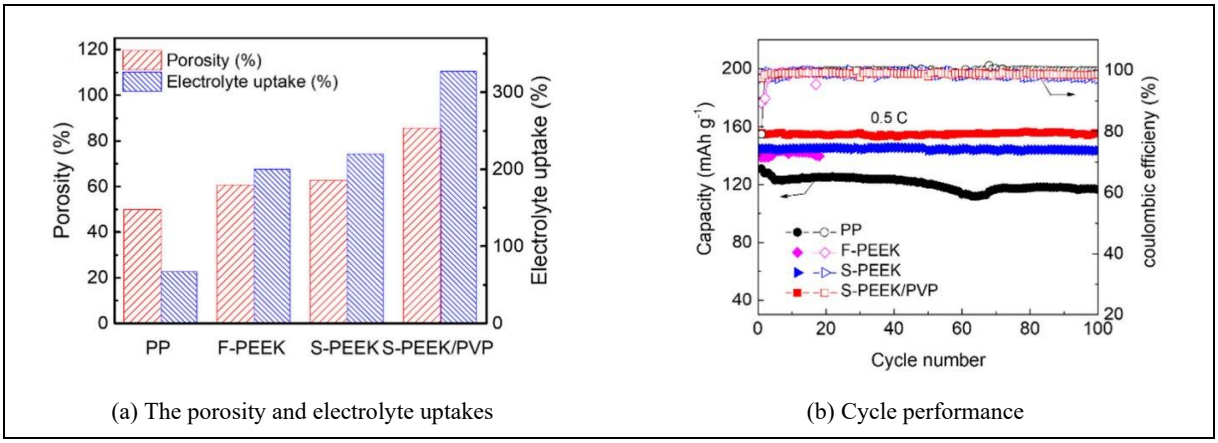
5.4. Energy and Electronics Applications

Tensile strength, chemical resistance, formability and thermal stability make PEEK-based materials popular in drilling and exploration. They also act as a barrier to metallic structures from gases and corrosive liquids during operation. Radio frequency identification (RFID) tags, used to track machinery during drilling and production, are also found on PEEK. Its great resistance to harsh conditions and corrosive environments also makes it ideal for sealing, connectors, and valve seats in subsea offshore oil and gas production systems. Inappropriate material selection can lead to catastrophic events such as oil spills, release of hazardous gas compounds and injuries and deaths. Excellent mechanical strength and resistance to changing temperatures, pressures, and weathering make VESTAKEEP® (AG, 2025), Evonik's commercial PEEK product, perfect for building subsea exploration equipment. Victrex® (plc, 2025) offers PEEK polymer products ranging from unfilled to carbon-filled and abrasion resistant. Solvay markets KetaSpire® (Polymers, 2025) PEEK, one of the best-performing semi-crystalline thermoplastics available on the market (de Leon et al., 2021).

PEEK's chemical resistance and insulation properties make it ideal for meeting the needs of the electronics and energy sectors. It is highly thermally stable for lithium-ion batteries (LIBs), portable devices, energy storage systems, and hybrid and electric cars. S-PEEK/PVP separators increase discharge capacity and cycling stability, making them ideal for energy storage applications. The design of the separator ensures high-temperature performance in lithium-ion batteries. According to studies, ideal membranes can maintain the capacity and stability required for applications in the energy sector. The S-PEEK/PVP membrane showed significant porosity (86%), increasing electrolyte absorption up to 327% (Figure 11b), which improved electrolyte absorption. The figure shows that the battery with the improved membrane retained 98% capacity

after 150 cycles at room temperature (Figure 11b). The wide electrochemical window above 5 V confirms the electrochemical stability of the S-PEEK and S-PEEK/PVP membranes (Niu et al., 2022).

The growing demand for chips in consumer electronics, communications equipment, and automobiles is exacerbating the chip shortage trend. Complex chips, particularly in semiconductor manufacturing, are composed of silicon wafers, electronic components, photomasks, photoresists, targets, chemicals, and other materials (Figure 12). The biggest challenge in semiconductor manufacturing is contamination control. This means that as electronic components become smaller and more complex, their tolerance to contaminants decreases, and manufacturing environments become more challenging. Plastics are widely used in packaging, transportation, processing, pollution prevention, effective contamination management, and significant process improvement in semiconductors. Hony Engineering Plastics Co., Ltd. manufactures PEEK (polyetheretherketone) and PI (polyamide) finished products and profiles for various industries. Electronic insulation films, interconnect devices, and wafer carriers most commonly use PEEK materials. Front-end processing, processing and inspection, and back-end processing are beneficial in wafer manufacturing. A wide variety of electronic devices, such as computers, printed circuit boards, printers, LEDs, batteries, switches, connectors, and PEEK, are available.



(a) The porosity and electrolyte uptakes (b) Cycle performance

Figure 11. *The porosity and electrolyte uptakes and cycle performance of F-PEEK, S-PEEK, and S-PEEK/PVP separators (Niu et al., 2022).* ***a)*** *The porosity and electrolyte uptakes,* ***b)*** *Cycle performance*

Figure 12. *Applications of PEEK in the Semiconductor and Electronics Industry: Components and Uses (Limited, 2022).*

5.5. Radioactive Applications

PEEK-based composite materials have been studied extensively with regard to their performance in acting as a neutron shield with the inclusion of boron carbide (B4C) due to its capacity to capture neutrons. It has been found in reports that B4C/PEEK composite has high performance in the field of fast neutrons, and the performance increases with the inclusion of boron carbide, Figure 13 (H. Li, Guo, Tu, Chen, & Zeng, 2024; Wu et al., 2020a). The application of 3D printing technology has assisted in improving the production process of the composite so that low-porosity test samples can be prepared with improved thermal and mechanical properties. The thermal deformation temperature of 30% wt. boron carbide PEEK was found to be 308.4°C, showing its usability in high-temperature conditions (Wu et al., 2020a).

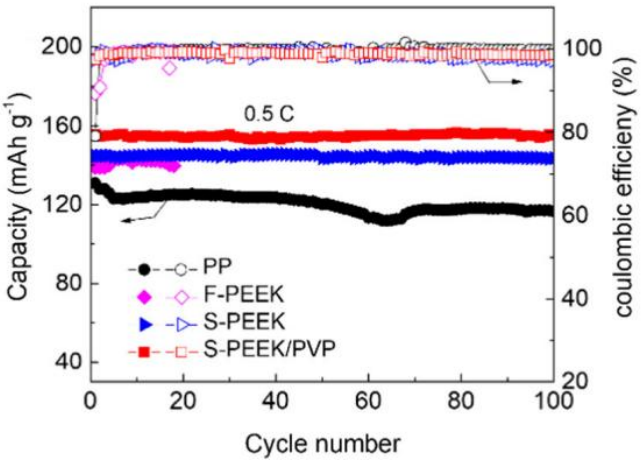
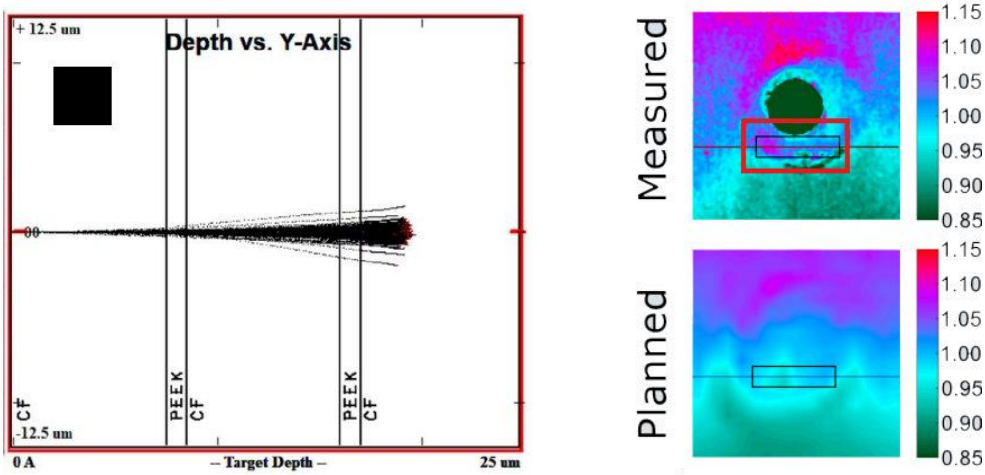


Figure 13. *Neutron protection curves of B4C/PEEK composite materials with different B4C addition amounts: (a) variation curves of neutron count and wavelength; (b) change curves of wavelength and protection effect (Wu et al., 2020a).*

Also, PEEK composite has been examined for gamma-ray protection with encouraging results in the tungsten-filled variants. The gamma-ray protection rates have a direct relationship with the percentage of used tungsten with higher bodies providing better protection, in particular from low-energy gamma rays from isotopes 137Cs compared to high-energy gamma rays from 60Co (Wu et al., 2020b).

In space applications, PEEK-based composites come in contact with radiation from protons, electrons, and high-energy particles. B4C/PEEK composites have been found to be effective in protection from such kinds of radiation, with Geant4 simulations confirming better performance with higher contents of boron carbide (H. Li et al., 2024; Wu et al., 2020a). Carbon fiber-reinforced PEEK (CF-PEEK) composites have also been examined in terms of high strength and thermal stability but with structural alteration due to exposure to high-energy ions such as He2+ influencing mechanical performance. Figure 14 shows penetration characteristics of He2+ ions at varying levels of energy in CF-PEEK composites, resulting in deeper penetration and local alteration in structure that can influence mechanical performance. It has been identified through research that ion irradiation may improve the hardness with alteration in microstructure, and thus there is a need to optimise CF-PEEK composites further to be used in space applications (Oromiehie et al., 2023).



(a) He^{2+} ion irradiation and SRIM calculations for CF-PEEK composites (Oromiehie et al., 2023).

(b) Dose Distribution Comparison of Nail Implants at 5 cm Depth (Laux et al., 2020)

Figure 14. *Comparative analysis of radiation effects on PEEK composites:* ***a)*** *He2+ ion irradiation and SRIM calculations for CF-PEEK composites (Oromiehie et al., 2023),* ***b)*** *Dose Distribution Comparison of Nail Implants at 5 cm Depth (Laux et al., 2020).*

PEEK has found application in radiotherapy in medicine, particularly in the production of radiolucent implants. The application of conventional metal implants can be a cause of dose inhomogeneities and imaging artefacts in radiotherapy and can undermine the efficacy of the treatment. Carbon fibre/polyether ether ketone (CF/PEEK) implants have been seen to cause minimal beam attenuation and artefacts and allow improved delivery of radiation to target volumes in contrast to conventional implants. It has been seen in studies that CF/PEEK implants produce less radiation attenuation (1%-2%) than titanium implants (5%-9%), Particularly in the nail model, where the measured dose was in closer agreement with the radiation planning with CF/PEEK implants than with titanium implants, Figure 14b, making them a better choice in radiotherapy applications (Laux et al., 2020).

There have been significant advances in the research on sulfonated poly(ether ether ketone) (SPEEK) and particularly in the application of SPEEK as a proton exchange membrane (PEM) in nuclear reactors in recent years. SPEEK has been found to be a suitable alternative to traditional Nafion® membranes due to its higher radiation stability and cost-effectiveness. It has been found through experiments that SPEEK films with a 70% degree of sulfonation have extremely high resistive capacity to high electron beam irradiation (up to 500 kGy), and thus they can be employed in situations where exposure to ionising radiation is high (Pajuste et al., 2024).

Also, the mechanical properties of such membranes have a loss in the yield strength with increased levels of sulfonation but an increase in the elongation at break, representing improved ductility required in a number of industrial applications. Thermal stability in SPEEK membranes is also found to be improved after irradiation, especially in those with low levels of sulfonation, which is crucial for applications where the material has to be resistant to high temperatures in addition to exposure to radiation. Figure 15 presents the TGA and DTA

traces of pristine and electron beam irradiated SPEEK membranes, indicating the thermal degradation behaviors and stability, and thus the suitability of SPEEK in high-temperature applications.

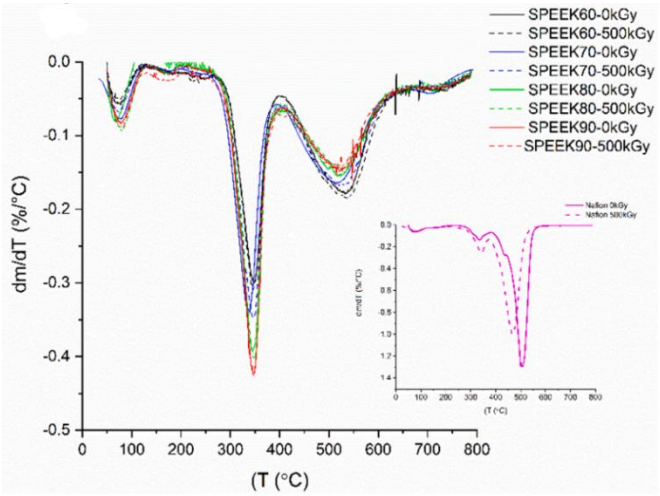


Figure 15. *DTA curve of pristine and electron beam irradiated SPEEK with DS 60–90% and Nafion® membranes (Pajuste et al., 2024).*

Table 11. *Comprehensive Applications of PEEK Across Industries.*

Industry	Key Applications	Material Properties Utilized	Citation
Medical/Biomedical	Implants, prosthetics, surgical instruments, orthopedic & dental implants, organ-on-a-chip devices	Biocompatibility, bone-like elastic modulus, radiation permeability, high thermal and mechanical stability, chemical resistance	(Sharma et al., 2021; Wang et al., 2022; Yang et al., 2025)
Organ-on-a-Chip	Microfluidic devices, biological sensing, diagnostic platforms	High thermal/mechanical stability, chemical/hydrolysis resistance, autohesive bonding capability	(Awaja et al., 2018; Song et al., 2018)
Marine	Underwater unmanned vehicles, seawater-lubricated bearings, tribological applications	Seawater resistance, wear resistance, self-lubrication, low friction, high strength	(Jacobs et al., 2005; Unal & Mimaroglu, 2006; Chen et al., 2011)
Aerospace	Aircraft components, turbine components, lightweight structural parts, bearings, gears	High-temperature resistance, lightweight, high strength, durability, chemical resistance	(Asyraf et al., 2022; Bearings, 2023)
Energy/Electronics	Lithium-ion battery separators, oil/gas components, semiconductor applications, electronic insulation	Chemical resistance, insulation, high-temperature resistance, electrochemical stability	(de Leon et al., 2021; Limited, 2022; Niu et al., 2022)

Table 11. *cıntinued*

Industry	Key Applications	Material Properties Utilized	Citation
Industrial/Manufacturing	Bearings, gears, high-temperature components, sealing elements	Self-lubrication, wear resistance, thermal/mechanical stability	(AG, 2025; plc, 2025; Polymers, 2025)
Radioactive	Neutron shielding materials, cable insulation, irradiation shielding, radiotherapy implants	Radiation resistance, neutron absorption, high-temperature resistance, radiolucency	(Wu et al., 2020a; Laux et al., 2020)
Nuclear	Proton exchange membranes, reactor components, radiation shielding	Radiation stability, thermal stability, mechanical durability, cost-effectiveness	(Pajuste et al., 2024)

5.6. Recycling of Polyetherketoneketone (PEKK) Composites

The spread of polyetherketoneketone (PEEK) into various sectors has created issues with waste disposal and sustainability. Recycling of PEEK composites has become crucial in order to reduce environmental impact and achieve a circular economy. As shown in Figure 16, fiber-reinforced polymer (FRP) composite recycling (Sachin et al., 2024) involves major methods such as mechanical, thermal, and chemical recycling, along with their industrial applications. Although PEEK composites present unique challenges, these established methods provide a valuable framework for comparison. The following section discusses potential recycling approaches, technical challenges, and industrial implementations based on recent literature.

Thermochemical recycling uses heat to break down composite materials by successfully melting and reforming PEEK composites. Experimentation confirms that processing under conditions of oxidation can improve fibre recovery, but with appropriate temperature control to prevent degradation of the polymer (Fontaine et al., 2021; H. Li & Englund, 2016). Thermal recycling of carbon fibre-reinforced PEEK composite was investigated to study the impact of varying atmospheres on fibre recovery and concluded total matrix degradation under oxidation conditions with risks of fibre oxidation. Pilot testing under conditions of 550°C revealed that only 42–46% of matrix extraction was achieved under conditions of inertia, while under conditions of oxidation, fibre decomposition occurred with steam interfering with decomposition (Fontaine et al., 2021).

Mechanical recycling entails comminuting composites into small pieces to reuse them, presenting a cost-effective and environmentally friendly alternative, yet with reduced mechanical attributes (Zeyrek et al., 2023; Yildirimoglu & Ozturk, 2024). Post-consumer PEEK composite materials showed a 33% reduction in flexural modulus when compared with virgin counterparts (Zeyrek et al., 2023). Recent studies on mechanical recycling of carbon fibre-reinforced PEKK composites, involving shredded aircraft waste, demonstrated preserved thermal attributes but significant reductions in tensile and flexural strengths (up to 50–60%) due to fibre misalignment and resin degradation (Zeyrek et al., 2023). Given structural similarities between PEKK and

PEEK, analogous outcomes are anticipated for carbon fiber-reinforced PEEK composites subjected to mechanical recycling.

Chemical recycling employs solvents or reactive chemicals to depolymerise the polymer matrix to yield recovered fibres. While PEEK stability discourages chemical recycling, technological breakthroughs with solvolysis technologies are future-looking (Morici & Dintcheva, 2022; M. Yuan et al., 2024). Recycling methods under review underscore solvolysis's promise to deliver high-value carbon fibres, however, it encounters challenges in terms of high inputs of energy and high-cost solvent requirements, hindering its widespread adoption (Morici & Dintcheva, 2022). One potential direction to overcome these challenges is illustrated in an integrated approach combining carbon fiber reclamation and additive manufacturing. This strategy not only improves economic feasibility but also promotes sustainability through circular material reuse (Liu et al., 2021).

Microwave-assisted pyrolysis (MAP) is a new methodology leveraging microwave energy to pyrolyse composite materials without degrading reinforcement fibre integrity. MAP has been demonstrated by studies to successfully generate high-grade carbon fibres from PEEK composites with minimal damage (Chen et al., 2022). Recycling of waste CFRP from bike components with MAP was found through studies to result in successful epoxy matrix pyrolysis, with no degradation of fibre integrity, having recovered fibres with enhanced heat stability and improved interfacial adhesion in polyamide.

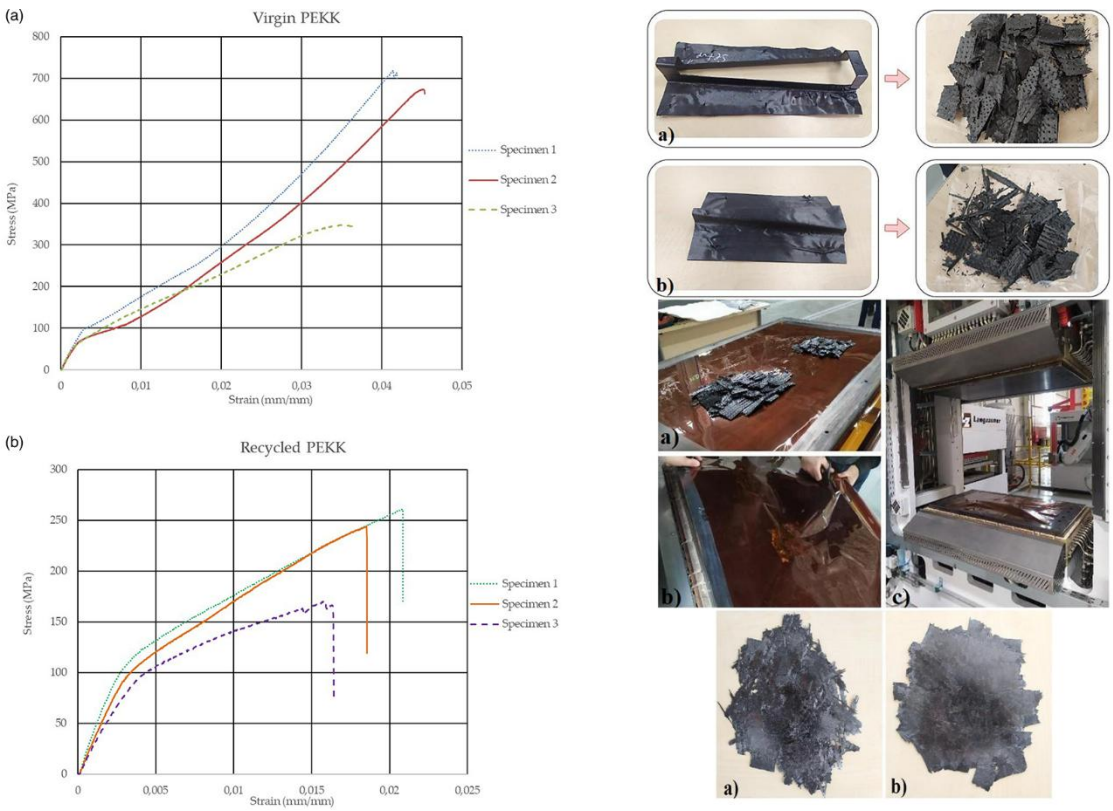


Figure 16. *Stress-Strain Behavior and Recycling Process of Virgin and Recycled CF/PEKK Composites (Zeyrek et al., 2023).*

5.7. Challenges in Recycling PEEK Composites

Even with diverse methods of recycling available, there are various challenges retarding broad adoption of PEEK composite recycling.

One major challenge lies in the high price of recycling processes. Sophisticated procedures like microwave-assisted pyrolysis and chemical recycling are energy-intensive and require high-cost equipment, thus becoming less economically favourable than generating new materials (M. Yuan et al., 2024; Vogiantzi & Tserpes, 2025).

One such challenge is mechanical property loss in recycled PEEK composites. The recycled composites, relative to virgin materials, also record reductions in flexural modulus and tensile strength, restricting their application in high-performance applications (Zeyrek et al., 2023; Yildirimoglu & Ozturk, 2024).

Some of these methods, including incineration and chemical processing, also have associated negative environmental impacts. They can lead to the release of toxic chemicals and greenhouse gas emissions, with issues of sustainability and compliance with regulations (Morici & Dintcheva, 2022; M. Yuan et al., 2024).

Moreover, its complex architecture poses a technical challenge too. The high adhesion between the reinforcement fibres and polymer matrix hampers effective separation and recovery of individual components upon recycling (Fontaine et al., 2021; H. Li & Englund, 2016).

Finally, there are no standardised methods of recycling, and this causes variations in quality and performance when PEEK composites are recycled. Without having procedures in place, comparing various methods of recycling and determining material reliability is difficult (Chatziparaskeva et al., 2022).

6. RESEARCH GAPS AND FUTURE DIRECTIONS

PEEK has been characterised by low outgassing and mechanical stability, but there are limited data available on its thermal performance when 3D printed under space conditions. Future studies must analyse the long-term influence of 3D printing on PEEK's mechanical performance and its thermally induced behaviours, as recent research shows potential thermally induced distortions critical to its structural integrity. Large-scale experiments in real-space conditions are needed to validate the performance and reliability of PEEK (Rinaldi et al., 2020). Most studies have concentrated on the mechanical, thermal, and electrical behaviour of PEEK under gamma irradiation, with too little consideration of its tribological behaviour, especially in nuclear contexts. It is imperative to understand irradiated PEEK's tribological processes, including surface degradation and internal crosslinking behaviour. Future research should address the long-term stability of PEEK in diverse conditions and how surface oxidation affects its mechanical properties. PEEK exhibits high resistance to gamma irradiation, with hardness and tensile strength showing minimal changes at doses up to 10 MGy (Chai et al., 2022).

6.1. Biomedical Research Priorities

The bioinert nature of PEEK remains a fundamental challenge for biomedical applications, restricting its bioactivity and necessitating continued research into improving osseointegration and bone remodelling. Recent advances in surface engineering, including hierarchical topographies and bioinspired coatings, have shown promise in overcoming these limitations (Sun et al., 2025; Zhang et al., 2025). However, long-term clinical studies are still needed to validate these approaches, particularly in dental applications where Alqutaibi et al. (2025) have identified key research priorities. Furthermore, understanding the long-term tribological behavior of PEEK composites under physiological conditions remains an important area for future investigation (S K, 2025).

Customisation of PEEK dental prosthetics through 3D printing and CAD/CAM presents opportunities for patient-specific implants, but variability in printer settings and the absence of standard protocols make this challenging. Interest in hybrid materials based on PEEK and bioceramics to improve biological interaction requires further evaluation. There is a critical need for clinical trials to determine the safety and efficacy of PEEK-based materials and 3D printing technology, addressing current gaps in regulatory evaluation (Taymour et al., 2024).

Although PEEK has great potential as a biomaterial for orthopaedic and dental implants, commercialisation is obstructed by its hydrophobicity and bio-inertness. Current investigations emphasise in vitro and short-term in vivo effects, leaving a significant gap in long-term human studies to establish PEEK's suitability as an implant material. More evidence is needed to demonstrate the efficacy of different surface treatments, and future research must explore the compatibility of various treatments as well as the potential of composites to enhance PEEK's osseointegration and bioactivity (Dondani et al., 2023).

6.2. Microfluidic and Lab-on-a-Chip Applications

The potential of PEEK in lab-on-a-chip (LOC) devices is substantial, yet several research gaps remain. Long-term stability data are not available, with current research addressing only temporary bonding performance. The long-term stability of PEEK-polystyrene devices under different conditions requires more exploration, and the scalability of fabrication methods remains a shortcoming. Detailed characterisation of interfacial properties between PEEK and PS, especially under plasma treatment, is needed to maximize bonding. The integration of PEEK with other sensors and polymers should be investigated in future studies to maximise device performance and to test PEEK-based LOC devices in real-world conditions (Awaja et al., 2018).

6.3. Materials Processing and Characterization

Research has predominantly investigated filler effects on PEEK's characteristics, yet the influence of changing melt viscosities remains unexplained. Adding short carbon fibers (SCFs) enhances mechanical and thermal stability; for example, 20 wt% SCF PEEK composites exhibit a tensile strength enhancement of approximately 63 MPa over neat PEEK. Future studies should explore the relationship between PEEK's molecular weight and

its corresponding viscosity and mechanical characteristics. Thorough investigations into the tribological behaviour of PEEK composites under various conditions are needed to determine wear mechanisms and evaluate long-term performance in real-world applications (F. Li, Hu, Hou, Hu, & Jiang, 2022).

7. CONCLUSION

Polyetheretherketone (PEEK) has become a cornerstone polymer in next-generation engineering and medical fields by benefiting from a unique combination of mechanical strength, heat stability, and chemical resistance. This overview emphasizes PEEK's versatility, spanning industrial requirements for high-performance components and energy storage systems through revolutionary biomedical applications in orthopedic implants and organ-on-a-chip devices. Improvements in polymer processing through nucleophilic aromatic substitution and cross-linking methods have optimized PEEK's thermal properties, allowing for customized solutions for demanding environments. Additive manufacturing (AM) technologies, e.g., fused filament fabrication (FFF) and digital light processing (DLP), further broaden their use by allowing for customized implants in addition to geometric complexities unattainable by traditional methods.

In biomedical applications, PEEK's elastic modulus (3–4 GPa), similar to human cortical bone, reduces stress shielding—a significant benefit over metal implants. Its bioinert nature, however, requires surface treatments such as plasma grafting and hydroxyapatite coatings for osseointegration. Advances in nanocomposites, such as carbon fibre- and graphene-reinforced PEEK, show impressive gains in tensile strength (up to 209 MPa) and wear resistance, putting these polymers in line for challenging tribology in marine and aerospace industries. Frontiers in the future show PEEK's versatility in integrating with extreme modalities, such as radiation-protected composites (e.g., boron carbide-PEEK) and in microfluidic systems.

Even with its merits, challenges endure. Mechanical recycling of PEEK composites continues to be economically and technically demanding, with up to 33% loss in flexural modulus. Stability under extreme radiation in the long term and standardized procedures for medical devices printed by 3D printing require study. Interdisciplinarity is key to further developing green manufacturing processes to streamlining PEEK’s performance in nuclear and space environments, where thermal stability, alongside resistance to radiation, is of primary interest. Scalable surface functionalization, in vivo bioactive composite studies, and integration of AI-assisted AM methods are future research needs for further fine-tuning microstructure control.

PEEK's path points the way in a new paradigm in material science, providing high-performance, environmentally friendly substitutes for established polymers and metals. By crossing current boundaries through advances in composite engineering and processing technology, PEEK has the potential to revolutionize use in regenerative medicine by standardizing 3D-printed PEEK implants, renewable energy, and more, cementing its position as a cornerstone of 21st-century material discovery.

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AUTHOR CONTRIBUTIONS

Aimen Motie Babaci: Conceptualization, Methodology, Investigation, Data Curation, Writing – original draft preparation. Nurbas: Supervision, Validation, Resources, Writing – review and editing. All human authors have read and legally accepted the final version under the CC BY-SA 4.0 framework.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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